



The selection of a research design





Research design

Is a plan and the procedures for research or it is a plan specifying the methods and strategies for collecting and analysis the information that we need in a research study. The plan involves several decisions and the overall decision involves which design should be used to study a topic.

❑ The choice of research design is based on

The research problem in the study.

The personal experiences of the researcher .

The audience for the study.



The three types of designs

- ❑ Qualitative research design
- ❑ Quantitative research design
- ❑ Mixed methods research.





❑ Qualitative research design

Is a means for exploring and understanding the meaning individuals or groups ascribe or human problem.



The process of research involves:

- Emerging questions and procedures,
- Data typically collected in the participant's setting.
- Data analysis inductively building from particular to general themes.
- The researcher making interpretations of the meaning data.
- The final written report has a flexible structure.



Quantitative research

Is a means for testing objective theories by examining the relationship among variables. These variable, in turn, can be measured, typically on instruments, so that numbered data can be analyzed using statistical procedures. The final written report has a set structure consisting of introduction, literature and theory, methods, results, and discussion.



Mixed methods research

Is an approach to inquiry that combines or associates both qualitative and quantitative forms. It involves philosophical assumptions, the use of qualitative and quantitative approaches, and the mixing of both approaches in a study..



The components that involved in research design

- Intersection of philosophy .
- Strategies of inquiry .
- Specific methods.





Philosophical worldviews

In writing about worldviews, a proposal might include a section that addresses the following :

- The philosophical worldview proposed in the study.
- A definition of basic consideration of that worldview.
- How the worldview shaped their approach to research.

Four different worldviews are :

- The post positivist Worldview
- Constructivism .
- Advocacy / participatory.
- pragmatism.





THE POSPOSITIVIST WORLDVIEW

The post positivist assumptions have represented the traditional form of research, and these assumptions hold true more for quantitative research than qualitative research . This worldview is called the scientific method or doing science research . It is also called positivist/postpositivist research, empirical science, and post positivism . It is called post positivism because it represent the thinking after positivism, challenging the traditional notion of the absolute truth of knowledge and recognizing that we can not be "positive" about claims of Knowledge when studying the behavior and actions of humans .



THE SOCIAL CONSTRUCTIVIST WORLDVIEW

Social constructivist is such a perspective, and it is seen as an approach to qualitative research .

Social constructivists hold assumptions that individuals seek understanding of the world in which they live and work .

Individuals develop subjective meanings of their experiences and these meanings are varied and multiple, leading the researcher to the complexity of views rather than narrowing meanings into a few categories or ideas.



THE SOCIAL CONSTRUCTIVIST WORLDVIEW

The goal of the research is to rely on participants' views of the situation being studied .

The questions become broad and general so that participant can construct the meaning of a situation, typically forged in discussions or interactions with other persons .



THE PRAGMATIC WORLDVIEW

Another position about worldviews comes from pragmatists and derives from the work of Peirce, James, Mead, and Dewey .

There are many forms of this philosophy, pragmatism as a worldview arises out of actions, situations, and consequences rather than antecedent conditions as in postpositivism .

Instead of focusing on methods, researchers emphasize the research problem and use all approaches available to understand the problem .



Pragmatism provides a philosophical basis for research :

- Pragmatism is not committed to any one system of philosophy and reality .
- Individuals researchers have a freedom of choice .
- Pragmatists do not see the world as an absolute unity .
- Truth is what works at the time . It is not based in duality between reality independent of the mind or within the mind
- The pragmatist researchers look to the '**what and how**' to research, based on the intended consequences – where they want to go with it
- Pragmatists have believed in an external world independent of the mind as well as that lodged in the mind



STRATEGIES OF INQUIRY

Strategies of inquiry are types of qualitative, quantitative, and mixed methods designs or models that provide specific direction for procedures in a research design . Others have called them approaches to inquiry or research methodologies



Strategies of inquiry include :

- Quantitative strategies
- Qualitative strategies
- Mixed methods strategies

Quantitative Strategies

Strategies of inquiry associated with quantitative research were those that invoked the post positivist worldview .

These include true experiments and less rigorous experiments called quasi-experiments and correlational studies and specific single-subject experiments .

These strategies have involved complex experiments with many variables and treatment .



- **Survey research** provides a quantitative or numeric description of trends, attitudes, or opinions of a population by studying a sample of that population . It includes cross- sectional and longitudinal studies .
- **Experimental research** seeks to determine if a specific treatment influences an outcome . This impact is assessed by providing specific treatment to one group and withholding it from another and then determining how both groups scored on an outcome .

Qualitative strategies

In qualitative strategies the numbers and types of approaches have become more clearly visible during the 1990s and into the 21st centuries .

- **Ethnography** is a strategy of inquiry in which the researcher studies an intact cultural group in a natural setting over a prolonged period of time by collecting, primarily, observational, and interview data .
- **Grounded theory** is a strategy of inquiry in which the researcher derives a general, abstract theory of process, action, or interaction grounded in the views of participants .



- **Phenomenological research** is a strategy of inquiry in which the researcher identifies the essence of human experiences about a phenomenon as described as described by participants .
- **Narrative research** is a strategy of inquiry in which the researcher studies the lives of individuals and ask one or more individuals to provide stories about their lives .

*Thank
you!*